

DATA QUALITY CHECK USING TABLEAU PREP (NETFLIX FAIR DATASET)

Dataset Title: Netflix Movies and TV Shows Dataset

Source: Kaggle (Public Dataset)

Description: The dataset contains information about movies and TV shows including title, director, cast, country, date added, release year, rating, duration, and genre.

Reason for Selection: The dataset was selected because it is publicly available, well-structured, and contains multiple real-world data quality issues suitable for analysis.

Identified Data Quality Issues:

1. Missing values in columns such as director, cast, and country.
2. Duplicate records based on show_id and duration.
3. Inconsistent format in duration column (minutes vs seasons).
4. Incorrect data types in release_year.
5. Irrelevant fields such as original cast and director columns after transformation.

Tableau Prep Cleaning Workflow:

1. Converted release_year to appropriate data type.
2. Removed null values from critical fields like show_id, duration and title.
3. Replaced missing values using IFNULL for director, cast, country, and rating.
4. Added Pivot table (Rows to Column) dividing the duration field into movies (Mins) and TV Shows (Seasons)
5. Removed irrelevant fields after transformation.
6. Renamed fields for better clarity.

Data Quality Dimensions Addressed:

Completeness: Missing values handled using IFNULL.

Consistency: Duration column standardized.

Validity: Incorrect formats corrected.

Uniqueness: Duplicate rows identified and removed.

Relevance: Unnecessary fields removed.

Integrity: Data relationships maintained after cleaning.

Summary:

The dataset initially contained multiple data quality issues such as missing values, inconsistent formats, and duplicate records.

Using Tableau Prep, a structured cleaning workflow was applied to improve the dataset.

The cleaned dataset is now more reliable, consistent, and ready for analysis.

Some limitations remain, such as dependency on original data accuracy and limited advanced validation.

